

Supporting Information for

Computational evolution of social norms in well-mixed and group-structured populations

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Supporting Information Text

1. Related literature

In the following, we review the theoretical literature on indirect reciprocity, and we discuss the relationship between our study and previous work. Since the previous literature is extensive, we focus on those studies that are most relevant to ours. For more comprehensive reviews that also discuss empirical work, see Refs. (1–6).

Previous models can be largely classified into two categories: public assessment models and private assessment models. Public assessment models assume that everyone agrees on any population member's reputation. In particular, in such models, it makes sense to speak of *the* reputation of player j; one could refer to that reputation by the variable m_j. Because there are no disagreements in how people assign reputations, models of public assessment are comparably easy to analyze. Private assessment models, on the other hand, assume that reputations (or opinions) may differ depending on the observer. In such models, player j's reputation depends on who you ask. Now, j's reputation from the viewpoint of player i would be encoded by the variable m_{ij}. This makes the mathematical analysis more intricate. In the following, we review each of the two categories separately.

A. Public assessment models.

Base model. We begin by introducing the seminal work by Ohtsuki and Iwasa (7, 8), which has been a foundation for many of the studies reviewed below. In their model, individuals in a large population repeatedly play donation games with different partners. At each time step, two individuals are randomly chosen to play a donation game, one as a donor and the other as a recipient. The donor decides whether to provide a benefit to the recipient at a personal cost. Each player has a reputation. Reputations are binary variables, indicating whether the player is good or bad. The donor's decision may be contingent on the reputations of the donor and the recipient. After this donation game, the reputation of the donor is updated depending on the donor's action and on the reputations of the two players. The above process is repeated indefinitely. We are interested in the long-term outcome of this process.

Whether cooperation can evolve in this model depends on how the donor determines its action and how reputations are updated. The rule that tells the donor whether to cooperate or to defect is called an "action rule." An action rule $P(X,Y)$ is defined as a map $P : \{G,B\} \times \{G,B\} \to \{C,D\}$. Here, X and Y are the reputations of the donor and the recipient. The rule to update reputations is called an "assessment rule." An assessment rule $R(X,Y,A)$ is defined as a map $R : \{G,B\} \times \{G,B\} \times \{C,D\} \to \{G,B\}$, where A is the action of the donor. After the action of the donor, the reputation of the donor is updated according to the assessment rule. We refer to a combination of an action rule and an assessment rule as a "strategy", or a "social norm". The action rules and assessment rules introduced above are "third-order norms". This means that the norm depends on the reputation of the donor, the reputation of the recipient, and the action of the donor. This class contains subclasses of simpler norms, referred to as "second-order" and "first-order" norms. Second-order norms depend only on the reputation of the recipient and the action of the donor while ignoring the reputation of the donor. Thus the rules are of the form $P(Y)$ and $R(Y,A)$. First-order norms are the simplest class of norms that assess the donor only based on its actions, whose rules are of the form $P(Y)$ and $R(A)$.

Ohtsuki and Iwasa (7, 8) investigated all 2,080 distinct third-order norms. They found that only eight norms, called the "leading-eight," can support cooperation for all benefit-to-cost ratios. Communities using one of the leading-eight norms are fully cooperative in the limit of rare errors; moreover, these norms form strict Nash equilibria. Two of the leading-eight, L3 (Simple Standing) and L6 (Stern Judging, SJ) are second-order norms. The leading-eight have the following properties in common:

1. The norm is able to maintain mutual cooperation: $P(G,G) = C$ and $R(G,G,C) = G$.
2. The norm is able to single out defectors: $R(G,G,D) = B$.
3. Individuals with bad reputation can be punished without harm in reputation: $P(G,B) = D$ and $R(G,B,D) = G$.
4. Individuals can recover from a bad reputation by cooperating with good group members: $P(B,G) = C$ and $R(B,G,C) = G$.

An important assumption in this model is that the entire population agrees on the reputation of each individual. This "public assessment" assumption is key to making the analysis tractable. Because the population is large, and rare mutants have no influence on public opinions, we only need to consider mutants with a different action rule to identify the Nash equilibria. To be a Nash equilibrium, the social norm needs to prescribe the best action for each context (i.e., for each combination of reputations of the donor and the recipient). This best action is uniquely determined by comparing the long-term benefit of cooperating with its immediate cost.

This initial work by Ohtsuki and Iwasa (7, 8) studies the "local" stability of the cooperation. It describes whether or not an infinitesimal amount of mutants with a different action rule have a selective advantage. In (9), the authors study the "global" evolutionary dynamics using the replicator dynamics of three species, ALLC, ALLD, and the discriminator (DISC), defined on the simplex $S_3 = \{(x_1,x_2,x_3) \mid x_1 + x_2 + x_3 = 1, x_i \geq 0\}$. The results are consistent with the previous studies, showing that L3 and L6 stably promote cooperation.

In Ohtsuki and Iwasa's initial work, reputations are binary variables, good or bad. Although this is the simplest assumption, this dichotomy may not fully capture all aspects of real reputation systems. Berger studied first-order norms with more nuanced

reputations, showing that cooperation becomes more tolerant against implementation errors (10, 11). Tanabe and Masuda investigated second-order norms with ternary reputations G (good), N (neutral), and B (bad), assuming an ordinal relationship between these three reputations (12). A more comprehensive study with ternary reputations was conducted by Murase et al. (13). They studied all third-order norms without any assumptions on the ordinal relationship between the reputation values. From more than 33 billion candidates, they found more than 2 million norms that form a cooperative ESS. These results suggest that there are many more ways to promote cooperation than the leading eight.

While it is common to assume actions and assessments are determined deterministically, it is also possible to consider stochastic norms. Murase and Hilbe studied third-order stochastic rules (14). In addition, they allowed observers to update the reputation of both the donor and the recipient. Although the number of stochastic norms is uncountable, they theoretically derived necessary and sufficient conditions for cooperative ESS. This model contains the classical model of Ohtsuki and Iwasa as a special case. For this case, they show that in addition to the leading-eight, there is another class of cooperative ESS, termed the "secondary-sixteen" (see Table S1). Among the third-order norms with binary reputations, the leading-eight and the secondary-sixteen are the only cooperative ESS.

Extended models. The simple baseline model outlined above can be extended in various ways. One direction is to allow for games different from the simple donation game. In (15), the authors add another possible action of "costly punishment". Their study concludes that costly punishment does not play an important role for the evolution of cooperation. According to them, successful norms do not use costly punishment for most parameter values. Nakamura and Masuda studied a model where reputations are unknown with a certain probability (16). They considered social norms with additional prescriptions for when a player's reputation is unknown. Similarly, they study in-group favoritism with a model where the population is divided into groups and players use different rules for in-group members and out-group members (17, 18). Nakamura & Ohtsuki (19) study indirect reciprocity in general simultaneous games. Here, both of the two randomly picked players simultaneously choose an action. In (20), the model is extended such that interactions are publicly known with a certain probability. Players can condition their actions and assessments depending on whether interactions are public or private. All these studies consider second-order norms to identify the stable and cooperative norms with methods similar to (7, 8).

Ghang and Nowak study games with optional interactions (21). In their model, players may refuse to play. The game only takes place when both players agree to play. Clark et al. study a model with simultaneous games (22), as in (19). For a certain class of games, they show that a simple first-order norm called "Grim-K" forms a cooperative Nash equilibrium. Finally Frean and Marsland study an indirect-reciprocity model with local transactions (23). In this way, they find that the best norms are the ones that operate with token exchanges, similar to the exchange of money.

Competition between norms. In the above studies, cooperative ESS are identified from a set of candidates. There is another set of studies in which different norms compete in some evolutionary process. Chalub, Santos, and Pacheco conducted a series of studies on the evolution of norms based on a multilevel selection model. Here, the population is divided into groups and norms are selected at the group level (6, 24, 25). Each group has its own assessment rule and reputations are shared within the group. Thus, there are no disagreements on reputations. However, within each group, players may have different action rules. These different action rules can compete. On the other hand, there is another level of selection at the group level, where groups with different assessment rules compete with each other. The main conclusion is that L6 (Stern Judging) is the most successful norm. The same conclusion is supported by Santos et al. (26) for an even broader class of norms. These findings are interpreted as an indication that complex norms do not necessarily translate into success; already the relatively simple norm L6 suffices.

A similar conclusion was obtained by Uchida and Sigmund (27). Instead of a multilevel selection model, they consider selection at the individual level. In their model, players may have different assessment rules. Thus, the reputation of each player may differ depending on the observer. However, we classify this model as a public assessment model because players having the same assessment rule share the same opinion about everyone, since there are no independent assessment errors. Based on a replicator equation describing the evolution of norms, they again conclude that L6 is the most successful norm.

Sasaki use replicator dynamics to compare the performance of a discriminating norm DISC against ALLC and ALLD (28). They suggest Staying (L7) is the most successful discriminating norm, because it leads to the largest basin of attraction towards cooperation.

Another line of research is the work conducted by Xu et al. (29). In their model, the donor's reputation is assigned by the recipient. This assigned reputation is then shared among the entire population without disagreement. Thus, even though players may have different assessment rules, this model is one with public assessments. They conduct an evolutionary analysis where both action and assessment rules are subject to change within the second-order strategy space. Surprisingly, they find no stable cooperation. Even L6 is susceptible to neutral invasion by mutants having different assessment rules. Even if such a mutant assigns a wrong reputation to the donor, which effectively works as an assessment error, both the mutant and the residents behave identically in the following interaction. This study highlights the difficulty of using public assessment models for the evolution of assessment rules.

Summary of public assessment models. Overall, we can summarize the previous work on public assessment as follows.

- Mathematically, public assessment models are easier to work with. For each context, it is straightforward to identify the best action. As a result, it is possible to identify all ESS. Among the deterministic third-order norms and for binary reputations, the leading-eight and the secondary-sixteen are the only strict Nash equilibria with full cooperation.

- The leading-eight have four simple principles in common. Similar principles also explain the cooperative Nash equilibria in models with more than two reputations, and with stochastic norms.

- When assessment rules are fixed, but action rules allowed to evolve, several studies find that L6 (Stern Judging) is the most successful norm.

B. Private assessment models. Public assessment models assume that each individual has a unique reputation, mutually agreed upon by all population members. This assumption may not always be realistic. In many cases, an individual's reputation may depend on which observer is asked. In private assessment models, individuals have their own opinions about each other. Such models typically assume that each population member independently observes the action of the donor with some probability q. Those who observe the interaction can then update the reputation they assign to the donor. Since assessments are subject to errors, individuals may have different opinions about the donor even if they observe the same interaction and apply the same norm. These disagreements make cooperation more difficult to maintain. They also make the mathematics more challenging.

In the public assessment model, it is possible to explicitly calculate the probability that an individual is regarded as good. Similarly, it is possible to derive the best action in each context, by comparing the long-term benefit of cooperating with its immediate cost. The number of possible contexts (such as when a good donor meets a bad recipient) is limited. In particular, when reputations are binary, there are only four contexts. The number of potential mutants is also often limited to those with different action rules.

In contrast, in private assessment models the number of possible configurations of the image matrix increases exponentially in the population's size. The number of potential mutants is also larger since mutants may now have a different assessment rule. These differences make it more difficult to study the evolutionary dynamics of norms under private assessments. One possible approach to obtain analytical results is to assume that the opinions of different observers are statistically independent. This is the case, for example, when each interaction is only observed by a single randomly chosen observer (30). Hence, this case is usually referred to as a model of 'solitary observation'. However, if interactions are observed by a sizeable fraction of individuals, and if assigned reputations depend on an individual's most recent actions, opinions of different observers will no longer be independent. To take these complexities into account, private assessment models generally need to be simulated.

In the following, we first review studies on the general model, followed by a discussion of solitary observation models. Lastly, we review work in which players can access the opinions of other players (e.g., through gossip or empathetic moral evaluation).

Base model (simultaneous observation model). To our knowledge, the first study to claim the crucial difference between the public assessment model and the private assessment model is by Takahashi and Mashima (31). They claimed that Shunning, which is not included in the leading eight strategies, can sustain cooperation when reputations are assigned privately. However, we surmise this result might be a consequence of the short duration of their simulations. Uchida et al. (32, 33) find that under private assessment, Stern Judging leads to a collapse of cooperation. Similarly, Oishi et al. (34) study how opinions are updated starting from a random initial configuration. They show that Stern Judging leads to a division of the population into two distinct groups. Each group assigns a bad reputation to out-group members, and individuals cooperate only within their own group. Yamamoto et al. (35, 36) study the evolution of social norms under a dynamics inspired by a genetic algorithm. Among the second-order social norms, they conclude that L3 (Simple Standing) is a keystone norm for cooperation. Although this study is similar to ours in the sense that they also consider the global evolutionary dynamics, they only consider second-order norms. Moreover, their simulation time is relatively short such that their conclusion may be sensitive to their assumptions on the all-good initial configuration. A similar work by Uchida (37) on the competition among second-order norms suggests that L6 (Stern Judging) is the key to cooperation. However, they assume there are no errors and the population starts from an all-good initial configuration. These assumptions effectively prevent the emergence of disagreements.

The leading-eight norms are comprehensively studied by Hilbe et al. (38). They consider a model with three possible strategies, allowing for ALLC, ALLD, and a leading-eight strategy. Their analysis suggests that none of the leading-eight can support cooperation in noisy environments with incomplete information. The study has been extended to include generosity (39), and to include quantitative assessments (40). While generosity does not seem to promote cooperation, quantitative assessments can help to maintain cooperation against ALLC and ALLD. A model that synthesizes direct and indirect reciprocity has been proposed in (41). There, strategies of indirect reciprocity are limited to first-order norms. Still, the authors identify a (stochastic) strategy called "Generous Scoring" that can sustain full cooperation in a Nash equilibrium.

While the above studies rely on simulations, Fujimoto and Ohtsuki take an analytical approach to explore the evolutionary stability of second-order norms (42, 43). They derive an approximation to describe the image matrix' time-evolution. In this way, they can infer conditions under which cooperation can be maintained. Their analysis shows that L3 (Simple Standing) can be evolutionarily stable. In contrast, L6 (Stern Judging) shows a poor performance. This approximation method can be extended to third-order norms, showing that the leading eight can be classified into several distinct types (44).

Lee *et al* follow a different approach (45–47). They consider a model in which reputations and actions are continuous variables between 0 and 1. By a linear approximation around the all-good states, they derive some necessary conditions for stable cooperation.

Actions and assessments are not always accurate and are subject to errors. In the models discussed, several types of errors are considered, which can significantly alter the respective conclusions. First, there are implementation errors, where a player may mistakenly defect when intending to cooperate, with probability ε_I. Provided that the error rate is small, implementation errors do not play a major role, because the two cases $\varepsilon_I \to 0^+$ and $\varepsilon = 0$ typically lead to the same results (43). This is because

even when an error causes a player to defect, all observers agree that a defection happened, and hence there is no disagreement. More significant are assessment errors, where the observer assigns a different reputation with probability ε. The introduction of such errors has a significant impact; specifically, the results for $\varepsilon = 0$ and $\varepsilon \to 0^+$ can be quite different since these errors cause disagreements among observers. Mathematically, however, such errors are often useful, because they ensure the respective system is ergodic (hence there is a unique stationary distribution). Among the studies mentioned, only (32, 34, 37) consider a case without assessment errors. However, (32) allows for imperfect observation, which can also lead to disagreements, and hence has a similar effect.

Solitary observation model. For general population compositions and parameters, the private assessment model can only be explored with simulations. However, the model becomes tractable if we assume that opinions of different observers are independent. There are different interpretations for how this independence may arise. One interpretation is that each interaction is only observed by a single individual (which roughly corresponds to a case with $q = 1/N$). Another interpretation is that each observer assesses a donor i based on a random sample of i's past interactions (not necessarily the most recently observed interaction). If the solitary observation model is simulated, simulations take slightly more time than in the simultaneous observation model, because the players' average reputations converge more slowly.

Okada et al. (3, 30, 48, 49) use solitary observation to study (mostly) second-order norms. They derive a replicator equation describing the evolution of ALLC, ALLD, and a discriminating action rule, for different assessment rules. According to their analysis, some norms such as L3 (Simple Standing) and L7 (Staying) can stabilize cooperation by coexisting with ALLC. On the other hand, Perret et al. (50) show that no homogeneous third-order population other than ALLD is an ESS.

Opinion-sharing models. The above studies suggest that cooperation is difficult to maintain in private assessment models. The main difficulty is that once different observers disagree on which reputations they assign to others, it becomes harder to enforce cooperative norms effectively. One solution to this problem is to allow players to synchronize their opinions. This requires that players have some means to learn the evaluations of other players. In this way, disagreements can be resolved, and the situation gets closer to the public assessment model.

Radzvilavicius et al. (51) study a model with empathy. In this model, an observer makes assessments based on the donor's opinion with probability E, and based on the observer's own opinion with probability $1 - E$. The more empathetic the players are (the larger E), the easier cooperation is to maintain. Public institutions can play a similar role for the evolution of cooperation (52). By taking into account the opinion of public institutions, individual disagreements can be resolved, again promoting cooperation. Kessinger *et al.* (53) study a model with gossip groups. Players in the same group share their opinions as in the public assessment model. Opinions between groups may differ. Kawakatsu *et al.* (54) propose another model of gossiping. They explicitly model the process of opinion exchange between players through gossiping. In this way, they derive a condition for how often gossiping needs to occur to promote cooperation. These four studies use the solitary observation model to derive a replicator equation when three strategies compete: ALLC, ALLD, and a discriminating strategy. Overall, they predict that Stern Judging is quite effective, as in the case of public assessment models. Similarly, pleasing behaviors also help to maintain cooperation (55). Here, players use the opinion of the majority (instead of their own opinion) to make a decision. This assumption again makes the situation more similar to the public assessment model.

Summary of the private assessment models. Overall, previous work on private assessment models can be summarized as follows.

- Compared to public assessment models, cooperation is harder to maintain because individuals might disagree on each other's reputation.
- These disagreements also make a mathematical treatment more difficult. While explicit formulas are feasible under the solitary observation assumption, more general cases need to be studied with simulations.
- The private assessment model is the more natural framework to explore the evolution of social norms. In this model, it is straightforward to consider individuals with different assessment and action rules. Many previous studies, however, only investigate the evolutionary dynamics among a limited number of strategies.
- Without opinion-sharing mechanisms, several studies claim that L3 (Simple Standing) is the most effective. L6 (Stern Judging), on the other hand, can no longer maintain cooperation. With opinion-sharing mechanisms, results are more similar to the public assessment model. In particular, L6 often prevails.

C. Contribution of our study. Our research on the private assessment model departs from the existing literature in several ways. Traditionally, evolutionary studies have focussed on a narrow range of strategies, such as first-order strategies (41), second-order strategies (30, 32–35, 42, 43, 48, 49, 51–53), or the leading-eight (38–40, 44, 55). Moreover, mutant strategies are typically taken from a restricted set, often only including ALLC and ALLD (30, 38–40, 44, 48, 49, 51–53). Such a restriction of possible mutant strategies bears the risk of being arbitrary. From the outset, it is not clear why ALLC and ALLD should be taken as the strongest competitors of any given norm. Conversely, a social norm's good performance against ALLC and ALLD does not guarantee that the norm would also perform well against others.

In contrast to these evolutionary studies, papers on evolutionary stability typically allow for a broader range of mutants (43–45, 50). However, those papers require mutants to be rare. As a result, these papers cannot explore the evolutionary transitions between different social norms.

Our study allows all strategies of a given complexity class (either all second-order strategies, or all third-order strategies). We do not only explore the local stability of a strategy, but also how a strategy can emerge and eventually take over the population. In a sense, our model is a quite natural one to consider. We neither restrict the set of possible mutants, nor how abundant mutants are allowed to be. Such a general analysis has been deemed infeasible, because of the enormous amount of simulations necessary. By running these simulations in parallel and on a supercomputer, we gain a more broad and robust understanding of the possible evolutionary pathways from one social norm to another.

2. Supplementary Results

In this section, we present some further results that were not included in the main text.

Figure S1 shows the cooperation levels at the selection-mutation equilibrium for the three-strategy system of ALLC, ALLD, and a leading-eight strategy. The equilibrium cooperation levels are calculated as $p_c(X)f_{\rm eq}^{\Delta}(X) + f_{\rm eq}^{\rm ALLC}$. Here, $p_c(X)$ is the self-cooperation level of the leading-eight strategy X, $f_{\rm eq}^{\Delta}(X)$ is the abundance of X in the selection-mutation equilibrium, and $f_{\rm eq}^{\rm ALLC}$ is the respective abundance of ALLC. Only three leading-eight strategies achieve notable cooperation levels, L1, L3, and L7. Even for these three strategies, cooperation is vulnerable to errors.

Figure S2 shows the same plot for the three-strategy system of ALLC, ALLD, and each secondary-sixteen strategy. None of the secondary-sixteen can maintain cooperation, irrespective of the benefit b or the error rate ε.

Figure S3 illustrates how cooperation levels among the third-order strategies $\mathcal{S}_3$ depend on various parameters for (left) well-mixed populations and (right) group-structured populations. Figure S3(A,B) depicts the effect of population size N. Over the range of tested population sizes, all qualitative results are robust. Figure S3(C,D) illustrates the effect of the probability ε_I to commit an implementation error. An implementation error occurs when a donor intends to cooperate but mistakenly defects. Interestingly, implementation errors slightly improve cooperation in well-mixed populations. However, they do not have a similarly positive effect in group-structured populations. Figure S3(E,F) depicts the effect of different observation probabilities q. Again, results remain qualitatively unchanged. Finally, Figure S3(G,H) shows the effect of different selection strengths σ. For group-structured populations, the same selection strength is used for both in-group and out-group imitations. When $\sigma \to 0$, selection is neutral. Here, all strategies are equally abundant, and the cooperation level reaches its neutral level of $1/2$. As σ increases, the cooperation level drops for well-mixed populations but increases for group-structured populations. Overall, these results suggest that the conclusions in the main text are robust with respect to parameter changes. The baseline parameters are $N = 50$, $\varepsilon_I = 0$, $q = 1$, and $\sigma = 1$. The other parameters remain the same as those in the main text.

Figure S4A compares the payoffs when all players of a population either adopt L1 or L2. Irrespective of the exact population composition, we find that L1 players have the larger payoff. A snapshot of the image matrix is shown in Figure S4B. There are $N = 50$ players, and half of them use L1 whereas the other half uses L2. The figure shows that L1 players are more likely to be assessed as good both by both L1 and L2, which helps to explain L1's payoff advantage.

Figure S5 shows which strategies are most likely to invade each of the leading-eight strategies. The plot is analogous to Figures 3E and 3F in the main text. The leading-eight are classified into three classes: Type I, Type II, and Type III. Strategies in the same class show similar invasion profiles (An exception is Type III, but Type III strategies generally do not promote cooperation).

Figure S6 repeats the analysis of Figure 4D in the main text, but with different parameters b and ε. Similarly, Figure S7 shows analogous figures to Figure 4E in the main text. Results are overall similar to the main text figures, suggesting that the respective findings are robust with respect to changes in the benefit and the error rate.

Finally, Figure S8 shows the results of Monte Carlo (MC) simulations for group-structured populations when the number of groups is finite. We follow the same procedure as in (56, 57). The population is of size $N \cdot G$, with G groups of size N each. In the short term, the players' strategies are fixed. Players gain their long-term payoffs by playing donation games within their group. At a longer time scale, the players' strategies can change either by intra-group imitation, inter-group imitation, or mutation. When an intra-group imitation occurs, a randomly chosen player i imitates the strategy of another randomly chosen player j from the same group, with probability $\varphi_{i\to j}^{\rm in}$ as defined in the main text. When an inter-group imitation occurs, the role model j is randomly chosen from a different group, and imitated with probability $\varphi_{i\to j}^{\rm out}$. When a mutation occurs, the strategy of a randomly chosen player is replaced by a randomly chosen strategy in $\mathcal{S}_3$. We assume that intra-group imitation occurs much more often than inter-group imitations and mutations. As a result, each group is typically homogeneous.

To simulate this evolutionary process with finite G, we conduct Monte Carlo simulations as follows. At each Monte Carlo step, we randomly choose a focal group i of strategy $\mathbf{p}$, whose strategy is updated. A mutation occurs with probability r and an inter-group imitation occurs with probability $1 - r$, where r is the relative mutation rate. When a mutation occurs, we draw a random strategy $\mathbf{q}$ from $\mathcal{S}_3$ and replace the strategy of the focal group i with probability $\rho_{\mathbf{p}\to\mathbf{q}}$. When an inter-group imitation occurs, we randomly choose a model group j, whose strategy is $\mathbf{q}$. The strategy of i is replaced by $\mathbf{q}$ with probability $\varphi_{\mathbf{p}\to\mathbf{q}}^{\rm out}\rho_{\mathbf{p}\to\mathbf{q}}$. We repeat this procedure for 10^7 MC steps with initial 10^6 MC steps for equilibration. The cooperation level at any given point in time is defined as $p_c^{\rm all} = \sum_{i=1}^{G} p_c(X_i)/G$, where X_i is the strategy of the group i.

Figure S8A shows the average cooperation level, and the fraction of L1 groups, depending on the number of groups. When $G=1$, the population is well-mixed and the cooperation level is low. As G increases, the cooperation level quickly rises. In particular, it does not take a large number of groups to observe substantial cooperation levels. Figure S8B shows a typical evolutionary trajectory for $G=100$. L1 plays a crucial role in maintaining cooperation. L1 groups are predominant for most of the time, only occasionally interrupted by other strategies.

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	(G,G)			(G,B)			(B,G)			(B,B)		
	P	$R_1(C)$	$R_1(D)$	P	$R_1(C)$	$R_1(D)$	P	$R_1(C)$	$R_1(D)$	P	$R_1(C)$	$R_1(D)$
$S1$	1	1	0	0	**0**	1	0	**0**	1	**0**	**0**	**0**
$S2$	1	1	0	0	**0**	1	0	**0**	1	**0**	**0**	**1**
$S3$	1	1	0	0	**0**	1	0	**0**	1	**1**	**1**	**0**
$S4$	1	1	0	0	**0**	1	0	**0**	1	**0**	**1**	**1**
$S5$	1	1	0	0	**0**	1	0	**1**	1	**0**	**0**	**0**
$S6$	1	1	0	0	**0**	1	0	**1**	1	**0**	**0**	**1**
$S7$	1	1	0	0	**0**	1	0	**1**	1	**1**	**1**	**0**
$S8$	1	1	0	0	**0**	1	0	**1**	1	**0**	**1**	**1**
$S9$	1	1	0	0	**1**	1	0	**0**	1	**0**	**0**	**0**
$S10$	1	1	0	0	**1**	1	0	**0**	1	**0**	**0**	**1**
$S11$	1	1	0	0	**1**	1	0	**0**	1	**1**	**1**	**0**
$S12$	1	1	0	0	**1**	1	0	**0**	1	**0**	**1**	**1**
$S13$	1	1	0	0	**1**	1	0	**1**	1	**0**	**0**	**0**
$S14$	1	1	0	0	**1**	1	0	**1**	1	**0**	**0**	**1**
$S15$	1	1	0	0	**1**	1	0	**1**	1	**1**	**1**	**0**
$S16$ (Forgiver)	1	1	0	0	**1**	1	0	**1**	1	**0**	**1**	**1**
common	1	1	0	0	$*$	1	0	$*$	1	†	$*$	$*$

Table S1. The prescriptions of the secondary sixteen. The top row (X,Y) indicates the reputations of the donor and the recipient, respectively. For instance, (G,B) means the case where a good (G) donor meets a bad (B) recipient. The rules P, $R_1(C)$, $R_1(D)$ indicate the prescribed action, the assessment when cooperation (C) is observed, and the assessment when defection (D) is observed, respectively. The entries that are different from each other are highlighted in bold text. In the bottom row, the common prescriptions are shown. $P(B,B)$, indicated by †, is 1 if and only if $R_1(B,B,C)=1$ and $R_1(B,B,D)=0$. All these norms are CESS for $b/c>2$.

$\mathbf{p}\backslash\mathbf{q}$	L1	L2	L3	L4	L5	L6	L7	L8	ALLC	ALLD
L1	0.02	0.00	0.02	0.02	0.00	0.00	0.02	0.00	0.03	0.00
L2	**0.14**	0.02	**0.13**	**0.12**	0.02	0.00	**0.14**	0.00	0.00	0.00
L3	0.02	0.00	0.02	0.02	0.00	0.00	0.02	0.00	0.03	0.00
L4	0.02	0.00	0.02	0.02	0.00	0.00	0.02	0.00	0.03	0.00
L5	**0.12**	0.01	**0.13**	**0.13**	0.02	0.00	**0.12**	0.00	0.00	0.00
L6	**0.08**	**0.13**	**0.06**	**0.10**	**0.09**	0.02	**0.15**	**0.29**	0.00	**0.40**
L7	0.02	0.00	0.02	0.02	0.00	0.00	0.02	0.00	0.03	0.00
L8	**0.11**	**0.10**	0.02	0.01	0.01	0.00	**0.11**	0.02	0.00	**0.03**
ALLC	0.02	**0.08**	0.02	0.02	**0.08**	**0.31**	0.02	**0.33**	0.02	**0.67**
ALLD	**0.10**	**0.07**	0.00	0.00	0.00	0.00	**0.05**	0.00	0.00	0.02

Table S2. Fixation probabilities between the leading-eight strategies, ALLC, and ALLD. Rows indicate the resident and columns the mutant strategy. Transitions with an almost neutral fixation probability are underlined. Transitions with a selective advantage are indicated in bold. Parameters are $b=5$, $\varepsilon=0.01$ and $N=50$.

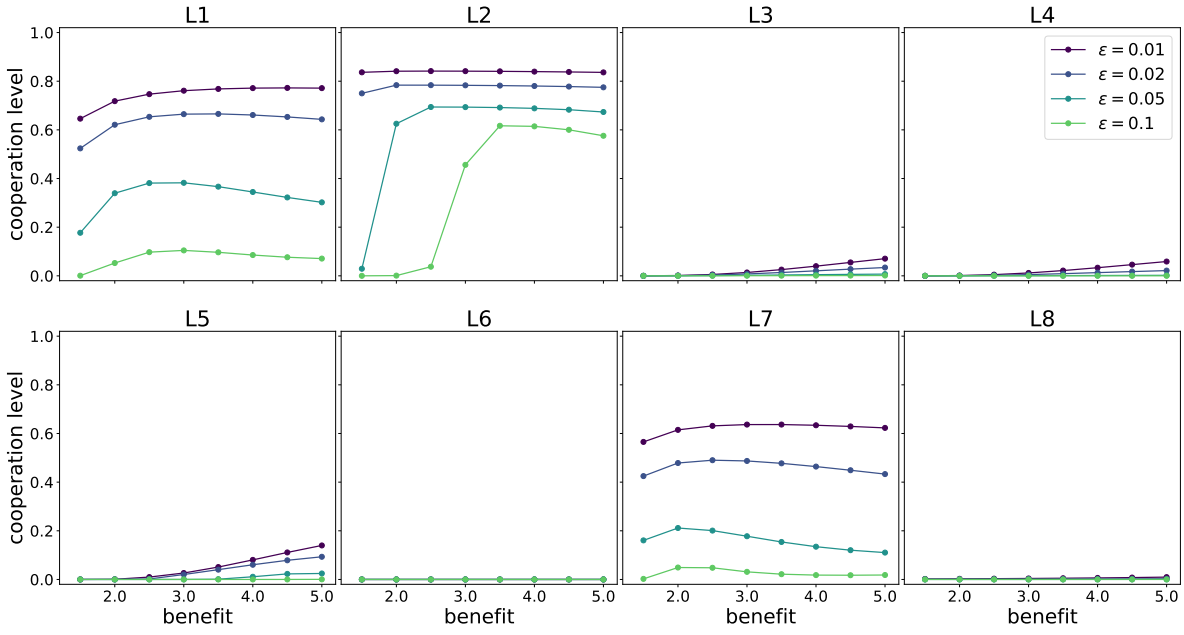


Fig. S1. Average cooperation levels when a leading-eight strategy competes with ALLC and ALLD. The horizontal axis shows the benefit of cooperation b. Each line in each panel shows the results for different error rates ε. The other parameters are the same as in the main text.

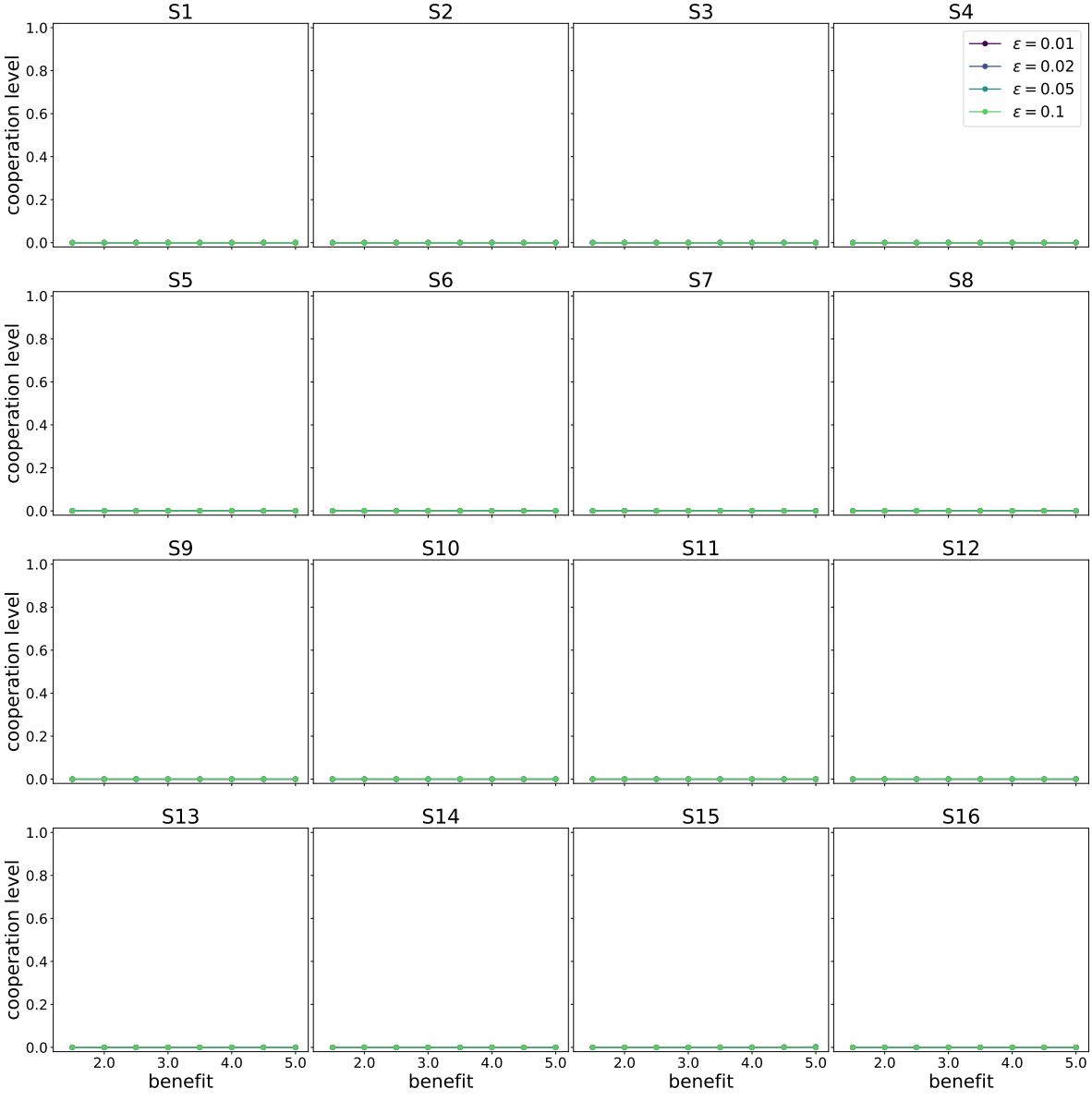


Fig. S2. Average cooperation levels when a secondary-sixteen strategy competes with ALLC and ALLD. The parameters not varied explicitly are the same as in the main text.

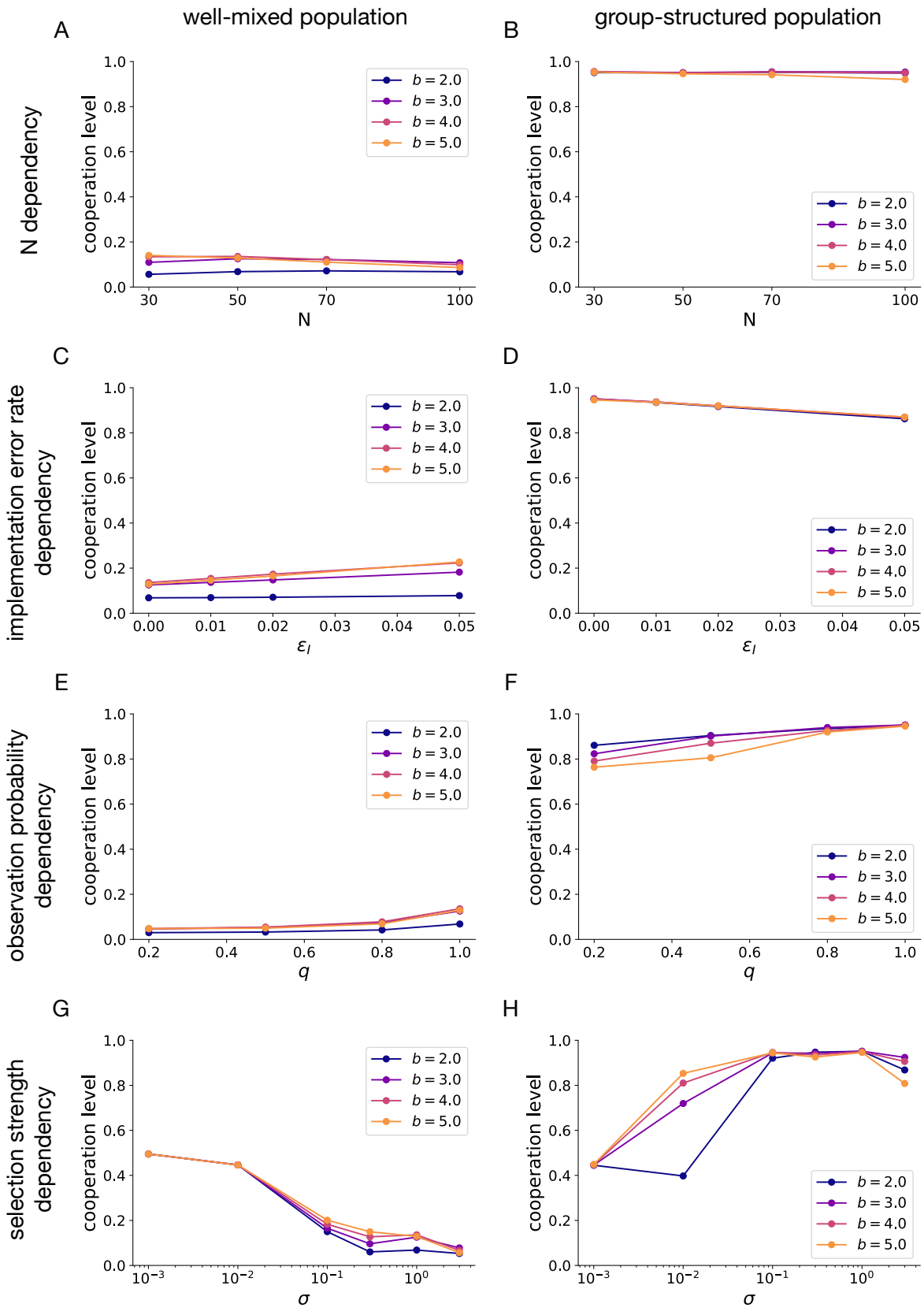


Fig. S3. Cooperation levels at the selection-mutation equilibrium in (left) well-mixed and (right) group-structured populations for different parameters. The different parameters we vary are (A,B) population size N; (C,D), the probability ε_I to commit an implementation error; (E,F) the observation probability q; and (G,H) the selection strength σ. For the assessment error rate, we usle $\varepsilon = 0.02$. The baseline parameters are the same as those in the main text.

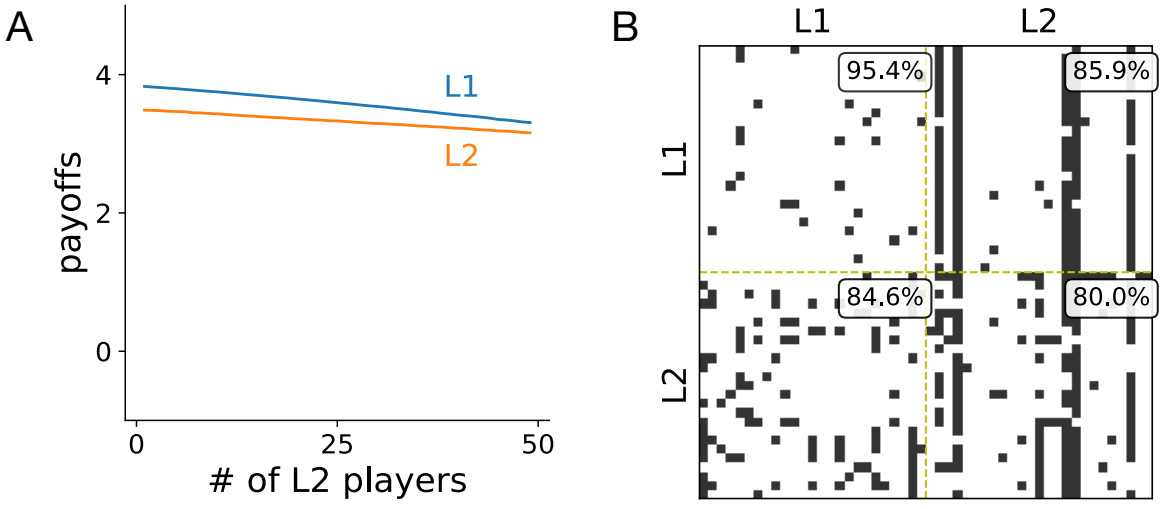


Fig. S4. (A) Payoff comparison when the population consists entirely of L1 and L2 players. For this graph, we set the benefit of cooperation to $b = 5$ and the error rate to $\varepsilon = 0.02$. The total population size is 50. The horizontal axis shows the number of L2 players. The remaining players adopt L1. For any population configuration, the payoff of L1 is greater than that of L2. This explains why L1 can selectively invade L2. Similar relationships are found for the other type I and type II strategies. (B) A snapshot of the image matrix $M = (m_{ij})$ when there are 25 players each for L1 and L2. The white and black squares indicate the good and bad opinions, respectively. The rows and columns indicate who is assessing and who is being assessed, respectively. The numbers inside the squares indicate the long-term average fraction of good opinions. L1 players are regarded as better than L2 players from the perspectives of both L1 and L2.

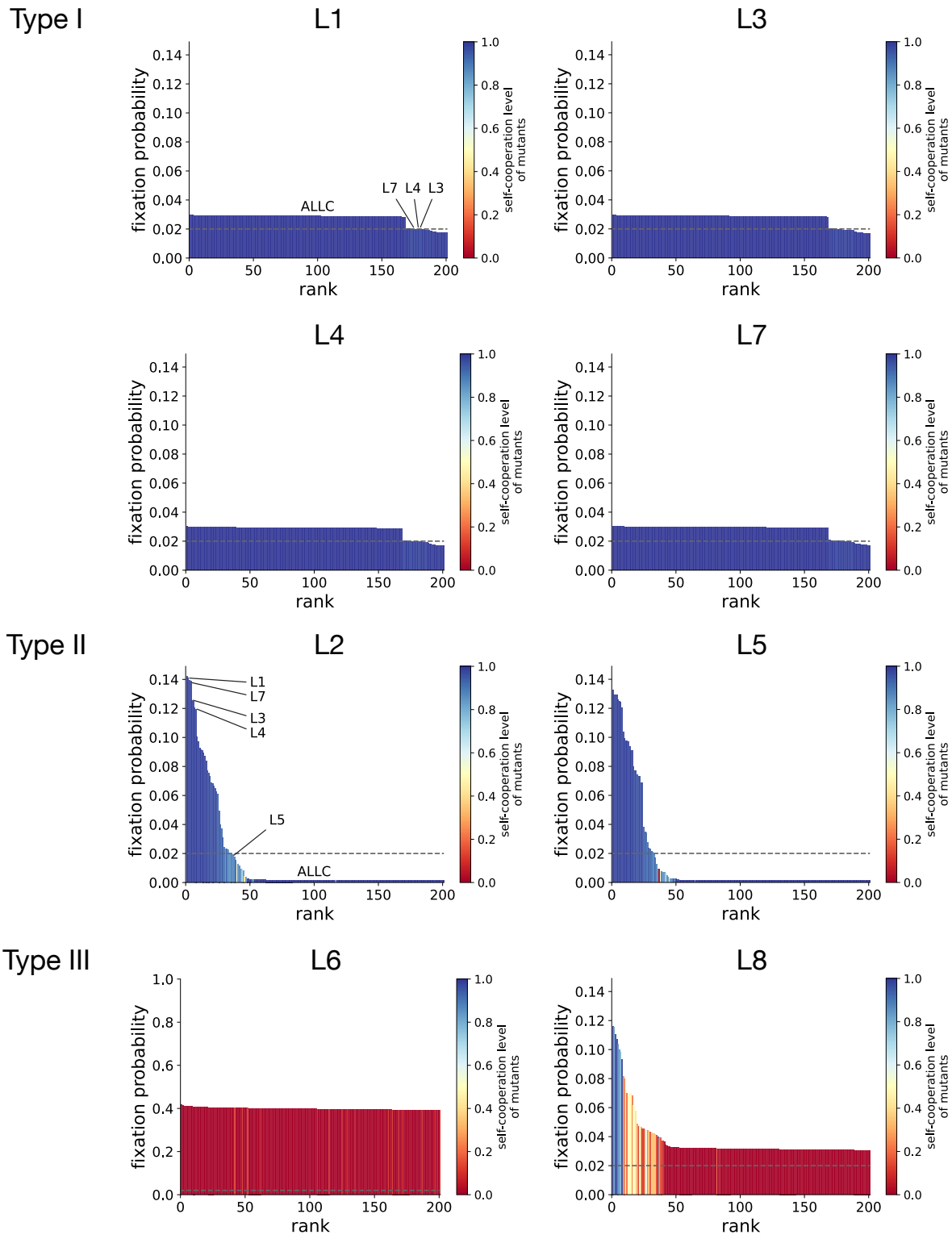


Fig. S5. Strategies that are most likely to invade each of the leading-eight strategies. The strategies $X \in \mathcal{S}_3$ are ranked according to their fixation probabilities against the leading eight residents. The panels are ordered according to the types of the leading-eight strategies. The results suggest that strategies belonging to the same type indeed behave similarly. Type I strategies (L1, L3, L4, L7) are most likely to be invaded by ALLC strategies. Type II strategies (L2, L5) can resist invasions by ALLC but are vulnerable to invasions by type I strategies. Type III strategies (L6, L8) are generally rather unstable. L6 is the most vulnerable strategy among the leading eight – note that here, the vertical axis is rescaled.

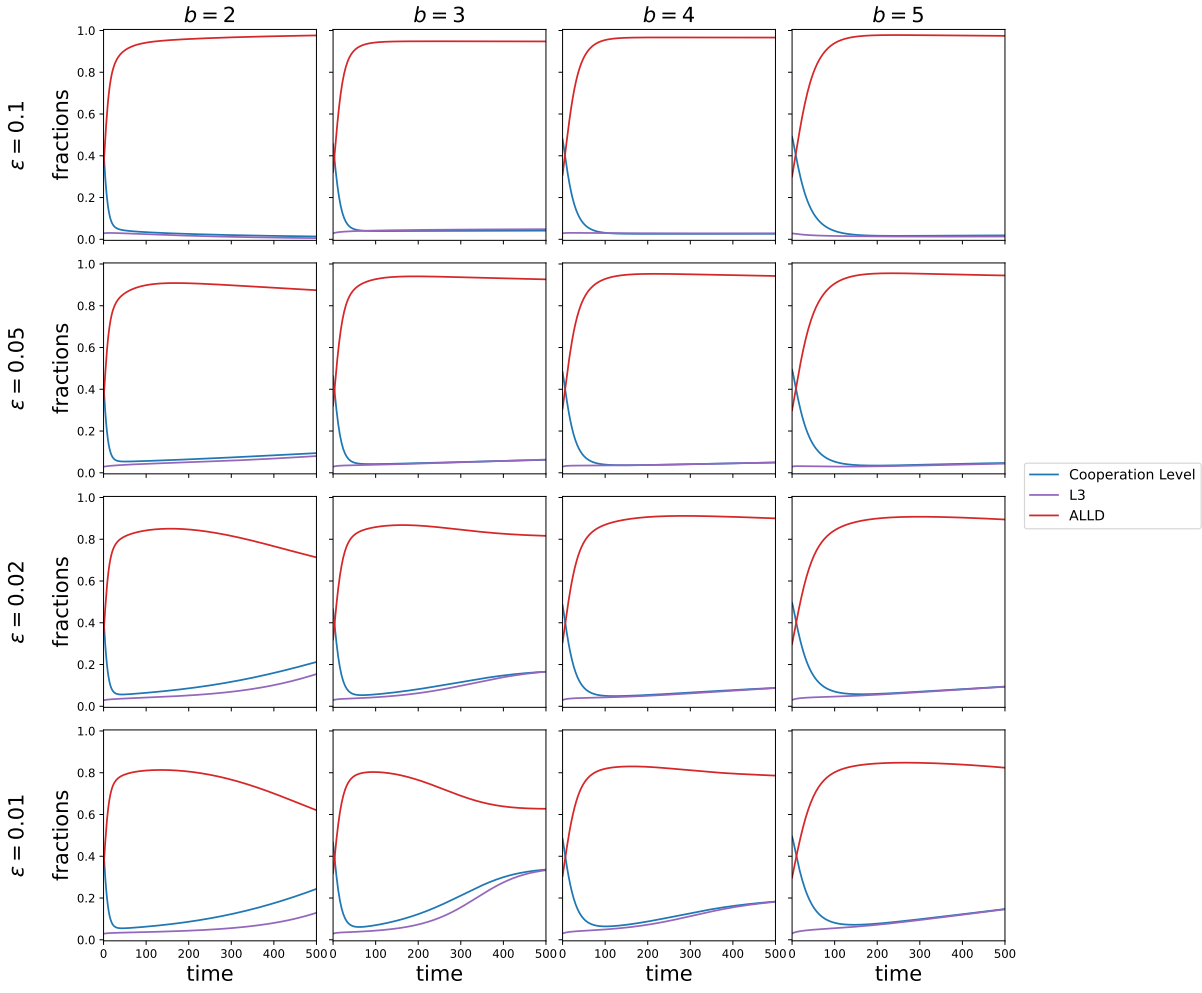


Fig. S6. Typical trajectories for the evolution of second-order strategies $\mathcal{S}_2$ in group-structured populations. As the figures show, ALLD is the most dominant strategy in all cases while L3 also evolves to some extent. However, in no case we observe the evolution of high cooperation levels.

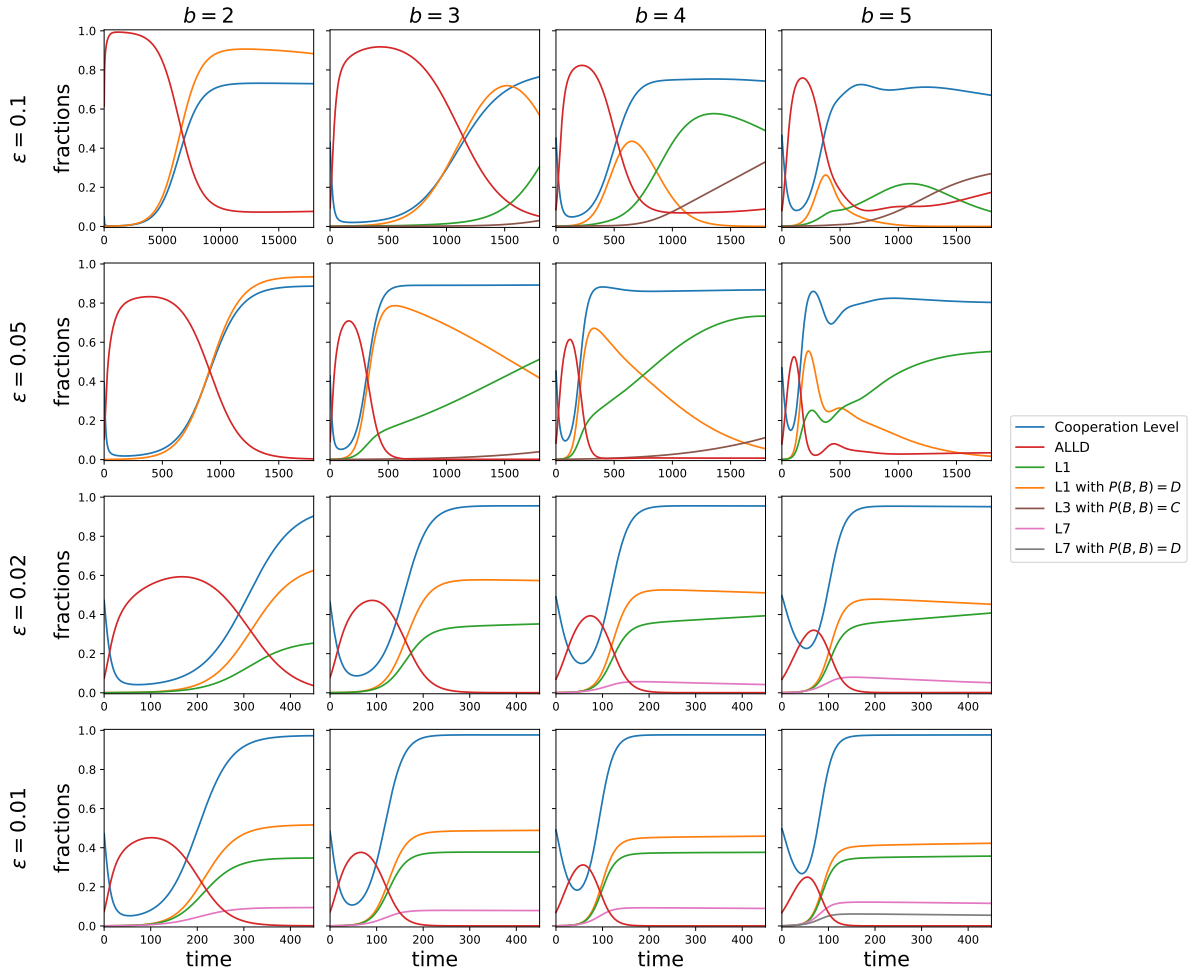


Fig. S7. Typical trajectories for the evolution of third-order strategies $\mathcal{S}_3$ in group-structured populations. Only strategies with an abundance of at least 5% are shown. Irrespective of the parameters b and ε, a similar pattern is found: ALLD grows initially. Then L1 with $P(B,B) = D$ takes over. At longer time scales, typically at $t \sim O(10^5)$, other type I strategies such as L1, L3, L7, and their variants increase via (almost) neutral drift. Which of these strategies dominates depends on the parameters. However, we systematically observe the evolution of type I strategies, leading to high cooperation levels.

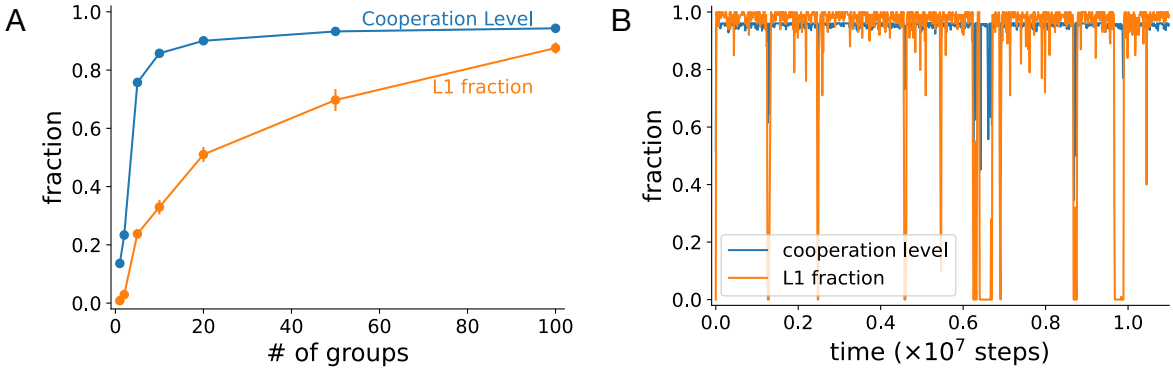


Fig. S8. Evolution among the third-order strategies in group-structured populations with finitely many groups. (A) We depict cooperation levels and the fractions of L1 at the selection-mutation equilibrium. When the number of groups is small, the cooperation level as well as the abundance of L1 is low. As the number of groups increases, cooperation levels as well as the abundance of L1 increases. Parameters here are $b = 3$ and $\varepsilon = 0.02$. We run the simulations for 10^7 MC steps with 10^6 MC steps for the initial transient. Results are averaged over 10 independent runs. Error bars indicate the standard errors. (B) A typical time series of the cooperation level and the fraction of L1 for $G = 100$, $b = 3$, and $\varepsilon = 0.02$.